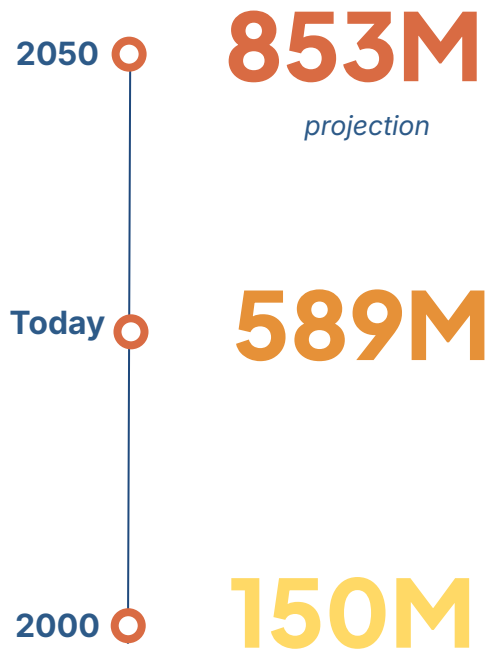


Diabetes Analysis

The global prevalence of diabetes has increased significantly worldwide



- Today, **11% adults (1 out of 9)** is living with the diabetes condition
- **90%** cases are **type 2 diabetes** driven by a combination of biological dysfunction, global environmental and lifestyle changes.

Identifying predictors for early diabetes detection

Pima Indian Dataset contains key clinical measurements for diabetes risks

Source

Kaggle (Khare, 2022)

Variables

8 Clinical predictors

Outcome (Diabetes/ Non-Diabetes)

Sample size

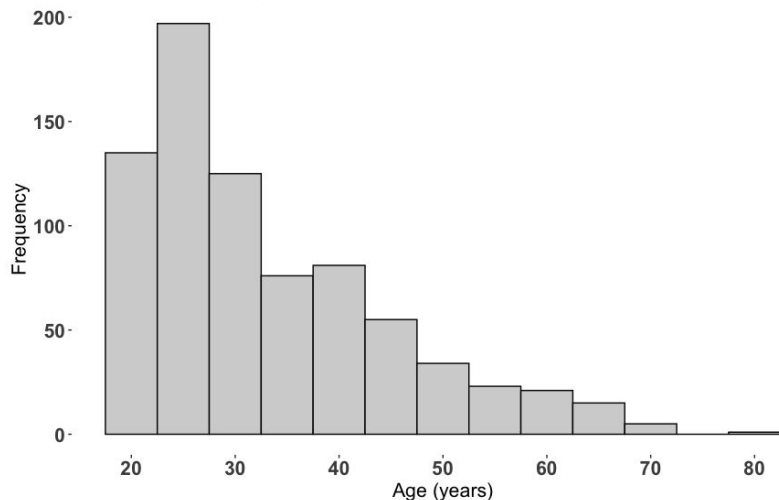
768 adults women
of Pima Indian Heritage

Quality limitation

Implausible zero values
in selected variables

The population has a diverse age, pregnancy history and diabetes status

Figure 1. Distribution of Age



Participant ranging from **21 to 81 years old**.

35% vs **65%**

Diabetes prevalence

85% vs **15%**

Women with ≥ 1 pregnancy

Clinical variables show wide variability and several outliers

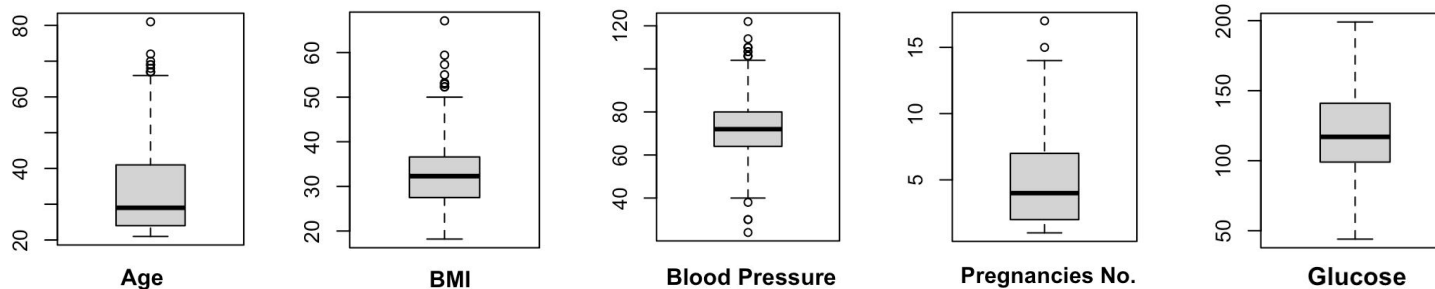
Table 1. Descriptive Statistics across clinical variables

Variable	Mean $\pm$ SD	Median	Range (Min - Max)	Mode
Age	33.24 $\pm$ 11.76	29.0	60.0 (81.0-21.0)	22 (f = 72)
BMI	32.46 $\pm$ 6.92	32.3	48.9 (67.10-18.20)	32 (f = 13)
Glucose level	121.7 $\pm$ 30.53	117.0	146.0 (199.0-44.0)	99, 100 (f = 17)
Blood pressure	72.41 $\pm$ 12.38	72.0	98.0 (122.0 - 24.0)	70 (f = 57)
Pregnancies No.	4.50 $\pm$ 3.22	4.0	16.0 (17.0 - 1.0)	1 (f = 135)

Glucose exhibits the highest variability, while average BMI (~32) indicates obesity across the sample.

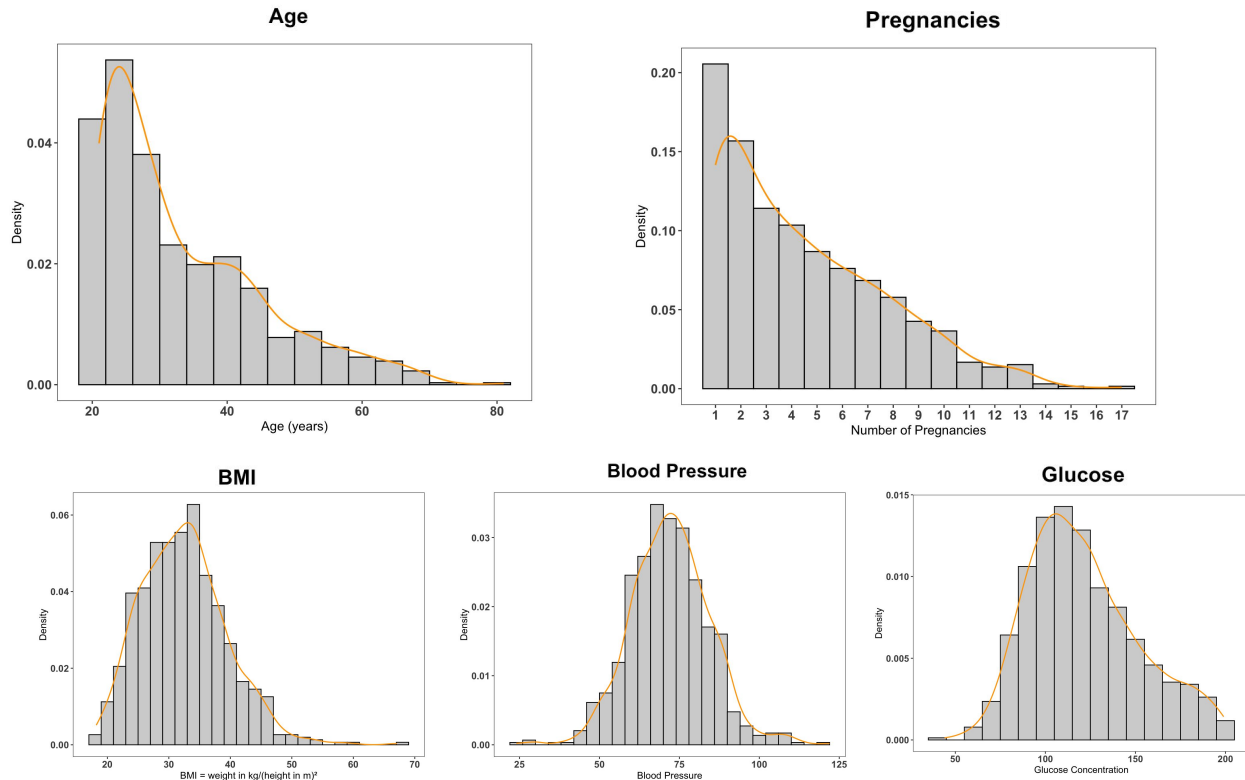
Centers for Disease Control and
Prevention, CDC (2024)

Figure 2. Distributions of clinical variables



Clinical variables show heterogeneous distributions

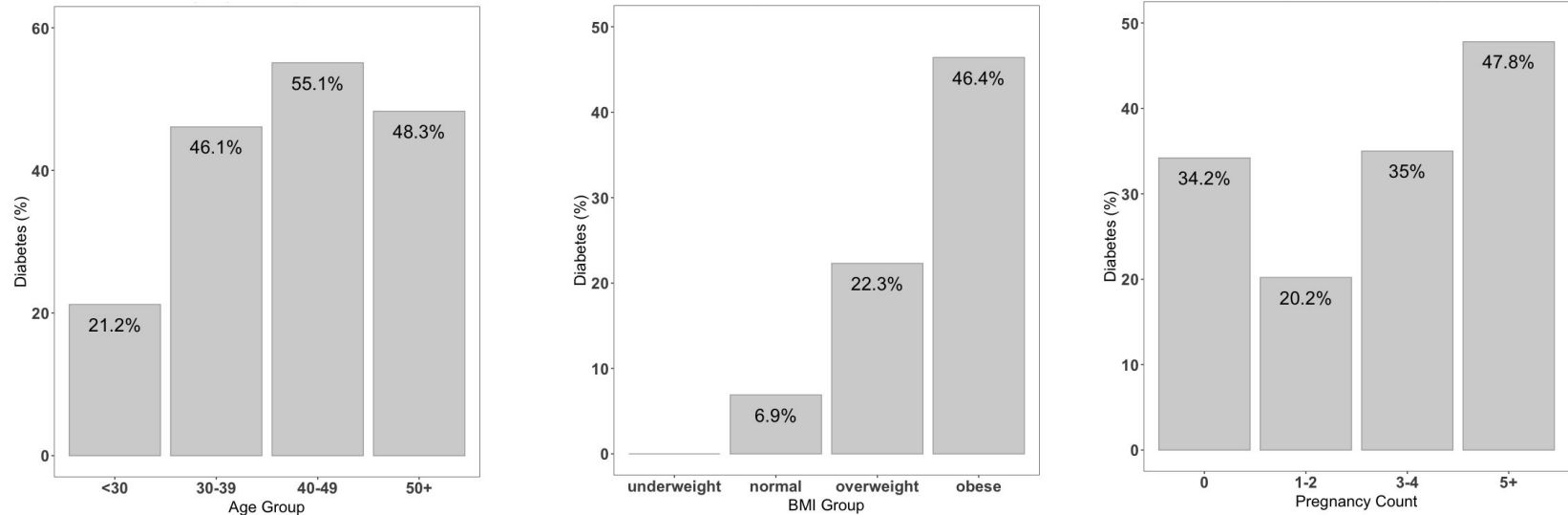
Figure 3. Distributions with density curve across clinical variables



Age and Pregnancies distribution are heavily right-skewed, whereas BMI, Glucose and Blood Pressure appear approximately symmetrical.

Diabetes rate increases across Age , BMI and Pregnancy groups

Figure 4. Diabetes rates by Age Group, BMI Group and Number of Pregnancies



Trends suggest potential association of Age, BMI and Pregnancy Group with diabetes risk.

Glucose level is higher for individuals with diabetes

Assumption check:

Shapiro-Wilk test: $p < 0.001$ both groups → **Normality violated**

Non-parametric test selected:

Mann-Whitney U test

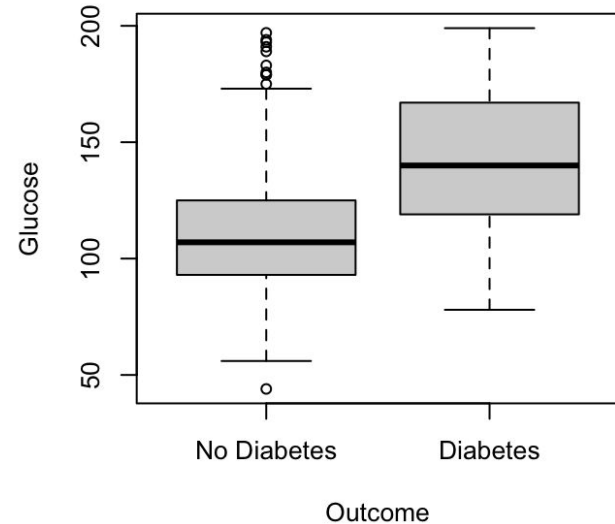
H_0 : There **IS NO difference** in the RANK SUMS of Glucose level between individuals with diabetes and non-diabetes.

H_A : There **IS a difference** in the RANK SUMS of Glucose level between individuals with diabetes and non-diabetes.

Test result: ($W = 27394$, $p < 0.001$) → H_0 rejected

Conclusion: Significant difference

Figure 5. Distribution of Glucose by Diabetes Outcome



Number of pregnancies is higher for individuals with diabetes

Assumption check:

Shapiro-Wilk test: $p < 0.001$ both groups → **Normality violated**

Non-parametric test selected:

Mann-Whitney U test

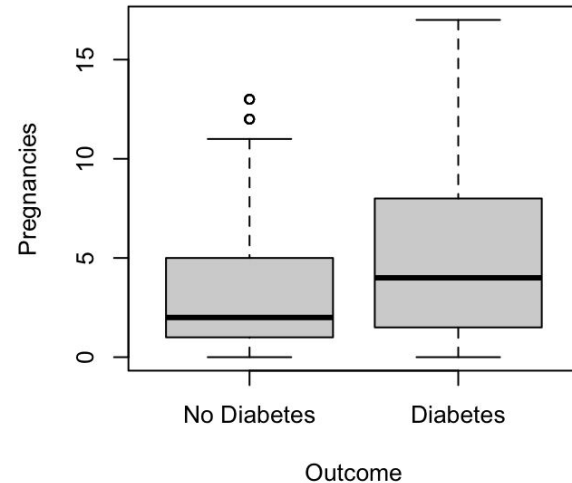
H_0 : There **IS NO difference** in the RANK SUMS of the number of Pregnancies between individuals with diabetes and non-diabetes.

H_A : There **IS a difference** in the RANK SUMS of the number of Pregnancies between individuals with diabetes and non-diabetes.

Test result: ($W = 50985$, $p < 0.001$) → H_0 rejected

Conclusion: Significant difference

Figure 6. Distribution of Pregnancies by Diabetes Outcome



Glucose levels increases across Age groups

Assumption check:

Shapiro-Wilk test: 2 out of 4 groups $p < 0.001$ → **Normality violated**

Non-parametric test selected:

Kruskal-Wallis test

H_0 : There **IS NO difference** in the RANK SUMS of Glucose level across different age groups.

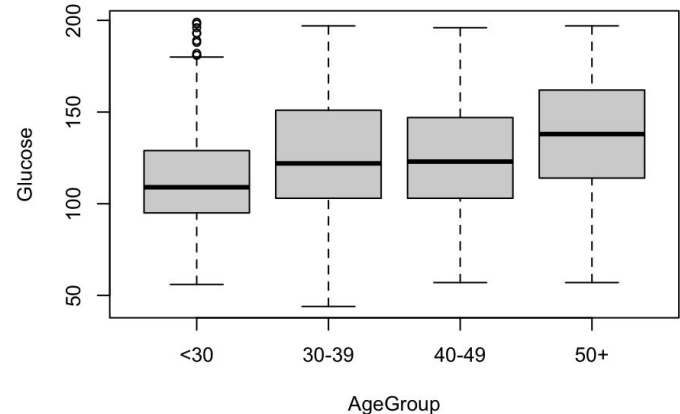
H_A : There **IS a difference** in the RANK SUMS of Glucose level across different age groups.

Test result: (Chi-sq = 59.042, df = 3, $p < 0.001$) → H_0 rejected

Conclusion: Significant difference

Post hoc Dunn's Test: 5 out of 6 pairs ($p < 0.001$)

Figure 7. Distribution of Glucose across Age groups



Diabetes rate increases across the Age groups

Assumption check:

Expected cell count (> 5) → **Assumption met**

Association test:

Pearson's Chi-square test

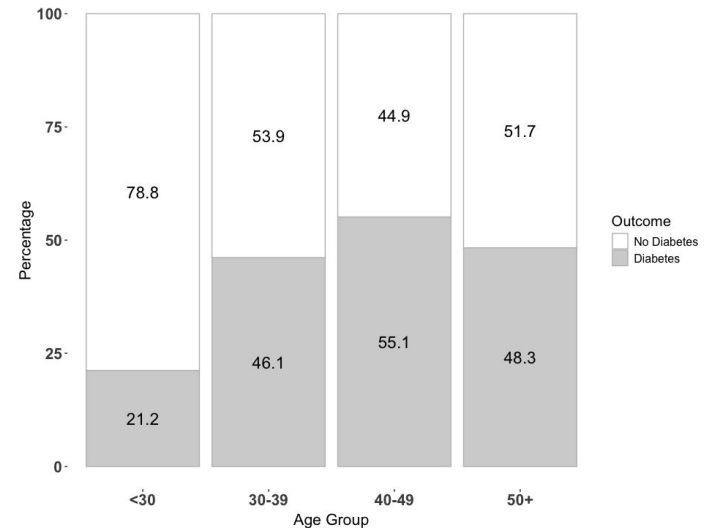
H_0 : There **IS NO association** between diabetes outcome and Age groups.

H_A : There **IS an association** between diabetes outcome and Age groups.

Test result: ($\chi^2 = 69.915$, $df = 3$, $p < 0.001$) → H_0 rejected

Conclusion: Significant association

Figure 8. Diabetes proportion across Age groups



Diabetes rate increases across the BMI groups

Assumption check:

Expected cell count (< 5) $\rightarrow$ **Assumption violated**

Association test:

Pearson's Chi-square test

H_0 : There **IS NO association** between diabetes outcome and BMI groups.

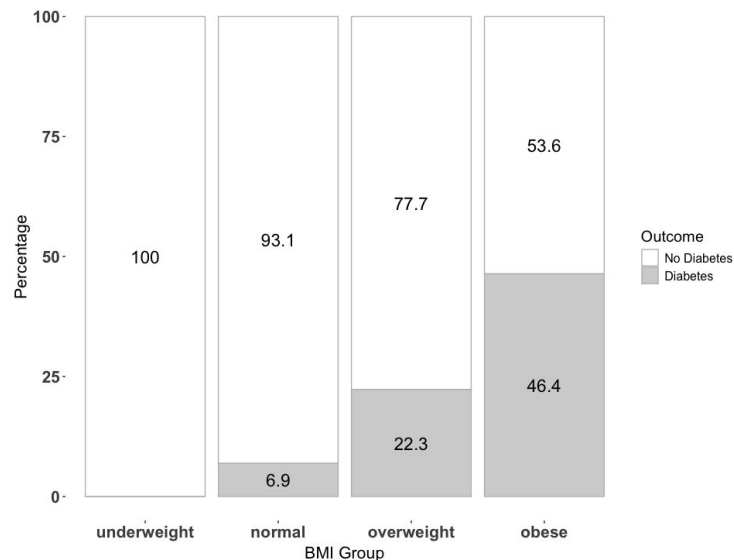
H_A : There **IS an association** between diabetes outcome and BMI groups.

Test result: ($\chi^2 = 77.057$ df = 3, $p < 0.001$) $\rightarrow H_0$ rejected

Fisher's Exact test: ($p < 0.001$) $\rightarrow H_0$ rejected

Conclusion: Significant association

Figure 9. Diabetes proportion across BMI groups



Most variables showed a positive correlation of weak to medium strength

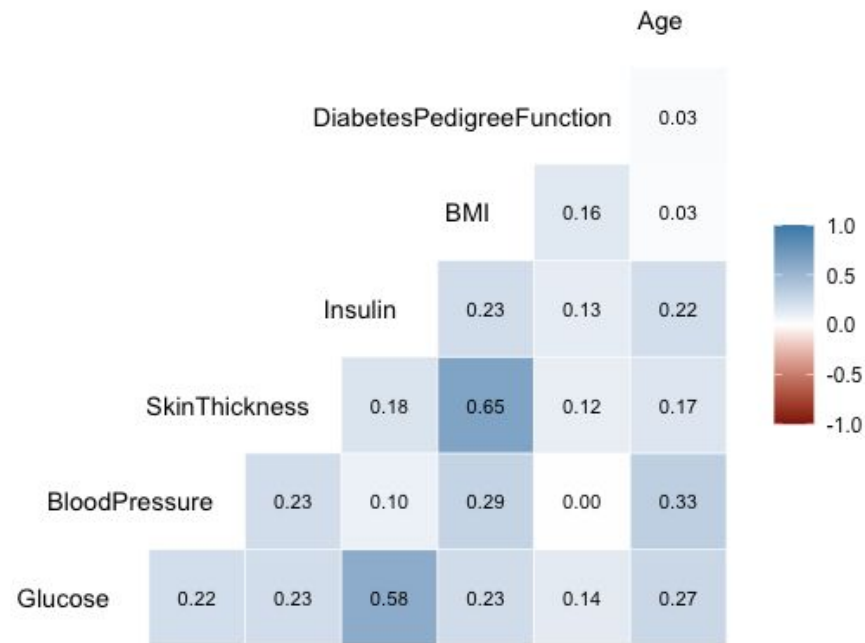
Figure 10. Correlation Matrix of continuous variables

A strong positive correlation was observed among

- Skin Thickness–BMI ($r = 0.65$)
- Glucose–Insulin ($r=0.58$)

Conclusion:

Further investigation is required to explore these positive relationships, as correlation does not imply causation.



Age, Blood Pressure and Insulin are primary predictors of Glucose level

Model Summary:

- Significant: $F(7, 384) = 36.76$, $p < 0.001$
- Explained variance: Adjusted $R^2 = 0.39$

Significant Coefficients:

- Age
- Insulin
- Blood Pressure

Table 2. Multiple Linear Regression Coefficients of Glucose

Predictors	Coefficient	t-Statistic	p-value	95 % CI
Intercept	60.03	7.112	<0.001	[43.3, 76.6]
BMI	0.130	0.535	0.593	[-0.35, 0.61]
Age	0.577	3.360	<0.001	[0.24, 0.92]
Pregnancies	0.074	0.141	0.888	[-0.95, 1.1]
BloodPressure	0.213	1.982	0.048	[0.002, 0.42]
SkinThickness	0.074	0.471	0.638	[-0.26, 0.38]
Insulin	0.133	12.293	<0.001	[0.11, 0.15]
DiabetesPedFun	4.179	1.153	0.250	[-2.95, 11.30]

Conclusion:

The Multiple Linear Regression model explained **39% of the variance** in Glucose level.

BMI, Age and Glucose predict diabetes risk

The Hosmer-Lemeshow Goodness-of-Fit test:

- Non-significant: $\chi^2(8) = 11.082$, $p = 0.197$
- No evidence of poor model fit ($p > 0.05$)

Model Classification:

- Accuracy: 77%
- Sensitivity: 57%
- Specificity: 88%

Conclusion:

The 'BMI + Age + Glucose' model demonstrated good fit and overall a good classification performance.

Table 3. Coefficients (BMI + Age + Glucose)

Predictors	Coefficient	p-value	OR	95% CI
BMI	0.090	< 0.001	1.094	[1.06, 1.12]
Age	0.029	0.000238	1.029	[1.01, 1.05]
Glucose	0.036	< 0.001	1.036	[1.03, 1.04]

BMI and Age interaction does not improve diabetes prediction

BMI + Age + Glucose

vs

Base Model + BMI x Age interaction

Likelihood Ratio Test:

- Non-significant: $\chi^2 (1) = 1.774$, $p = 0.183$
- Non-significant interaction between BMI and Age
- Base model retained

BMI + Age + Glucose + Pregnancy

vs

Extended Model + BMI x Age interaction

Likelihood Ratio Test:

- Non-significant: $\chi^2 (1) = 0.564$, $p = 0.453$
- Non-significant interaction between BMI and Age
- Extended model retained

Conclusion:

The interaction between BMI and Age did not significantly improved the models. The simpler models were retained.

Age independent effect diminishes once Pregnancy is included

Introducing Pregnancies into the model, led to Age becoming non-significant, suggesting Age's limited independent predictive contribution.

Conclusion: Further investigation is required to explore Pregnancies and Age relationship.

Table 4. Coefficients of the Extended Model

Predictor	Coefficient	p-value	OR	95% CI
BMI	0.091	< 0.001	1.095	[1.03, 1.12]
Glucose	0.036	< 0.001	1.030	[1.03, 1.04]
Pregnancies	0.109	< 0.001	1.116	[1.05, 1.19]
Age	0.012	0.194	1.012	[0.99, 1.03]

Conclusion

- Diabetes prevalence increases with Age, BMI category, and number of Pregnancies.
- Glucose and BMI emerge as the consistent independent predictors of diabetes risk.
- BMI and Age interaction does not improve the model fit, supporting the use of the simpler model.
- Age contribution diminishes once Pregnancy information is included.

Recommendation

- **Test the value of Pregnancies:** base (BMI + Age + Glucose) vs extended (BMI + Age + Glucose + Pregnancies)
- **Test the value of Age:** extended - Age (BMI + Glucose + Pregnancies) vs extended (BMI + Age + Glucose + Pregnancies)
- Validating the model on independent dataset and evaluating it on a diverse population to confirm generalisability.

e-portfolio: R code

https://agadjg.github.io/eportfolio/assets/scripts/diabetes_final_analysis.R

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